

Activity Guide — middle / module-14- magic-squares

IKS Curriculum

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Activity 1 — Class 3×3 Magic-Square Tournament

Module: Magic Squares — Bhadragaṇita · **Band:** Middle · **Duration:** 25 min

Purpose

Reinforce the day-by-day concepts of Magic Squares — Bhadragaṇita through a hands-on activity. Aligned with NEP 2020 §4.23 (experiential learning).

Materials

grids on paper, pencils, timer

What students do

In pairs, students race to construct the 3×3 magic square with 1–9 from a starting 9-position permutation. First pair with correct sums wins.

Step-by-step

1. **Setup (5 min)** — gather materials; explain the task.
2. **Working phase (15 min)** — students work individually or in pairs.
3. **Share (5 min)** — selected students share their results with the class.

Differentiation

- **Extension:** ask advanced students to do an additional iteration or analysis.
- **Support:** pair students who need scaffolds; offer partial templates.

Assessment

This activity is **formative**. The teacher observes participation, accuracy, and reflection quality. No grade is assigned.

Safety

- General safety briefing before any materials are handled.

- If herbs, plants, or food are involved: kitchen-amount only, no ingestion of unknown preparations.
- Outdoor activities: stay in the designated area; supervised by the teacher.

Activity 01 — Stair-Step Drill

Module: Magic Squares (*Bhadragaṇita*) · **Band:** Middle · **Day:** 4 · **Time:** 15 min

Type

Individual + paired practice

Goal

Each student constructs three 3×3 magic squares using Nārāyaṇa’s *turagagati* method, with the partner checking accuracy.

Materials per student

- 3 printed blank 3×3 grids
- The 5-step rule sheet (from Day 4 student-facing content)
- Pencil

Set-up (teacher, before class)

- Print blank 3×3 grids (3 per student).
- Have the 5-step rule visible on the board.

Instructions (student-facing handout)

Stair-Step Drill — work individually first, then in pairs.

Round 1 (5 min) — Slow build, rule sheet open. Construct a 3×3 magic square using the *turagagati* method. Take your time. Speak the steps out loud quietly. Verify all three row sums equal 15 before stopping.

Round 2 (4 min) — Faster, rule sheet open. Construct another 3×3 magic square from scratch. Time yourself.

Round 3 (3 min) — Rule sheet closed. Construct one more 3×3, this time from memory. Time yourself.

Pair check (3 min) — swap sheets with your partner. Did they get the same square? (They should — there is only one 3×3 magic square up to symmetry, and the algorithm always produces the same one.) Compare round-by-round times.

What to expect

All three squares should be identical:

2 | 7 | 6

9		5		1

4		3		8

If a student’s square is different, walk through their steps to find where the rule was misapplied — usually step 5 (“blocked → directly below”).

Discussion prompts (last 2 min)

- Did you get the same square all three times? If yes, why? If no, where did the difference come from?
- What did Round 3 (from memory) feel like compared to Round 1?

Looks-like / sounds-like

	Looks like	Sounds like
Going well	Student narrates “1 in top middle, then up-right, off the top so wrap to bottom...”	“Where does 6 go? Up-right from 5 — that’s outside the top, so wrap to...”
Needs intervention	Student stares at the blank grid; doesn’t know where to begin	Silence; “I don’t get it.”

Safety / sensitivity

- No timed competition between students. Each student times *themselves*.

Differentiation

- **Tier up:** Construct a 7×7 magic square using the same method. Constant should be 175. Verify two rows and the main diagonal.
- **Tier down:** Stay on the 3×3. The rule must be automatic before scaling up.

Clean-up

- All 3 constructed squares stay in the student workbook for later reference.

Activity 02 — Cross-cultural Comparison

Module: Magic Squares (*Bhadraganita*) · **Band:** Middle · **Day:** 8 · **Time:** 15 min

Type

Paired worksheet with discussion

Goal

Pairs compare three magic squares from three different cultures and identify the structural features they share or differ on.

Materials per pair

- 1 comparison worksheet (template below)
- Printed images of the three squares (or projected on the board): Lo Shu, Khajuraho, Dürer

Instructions (student-facing)

Cross-Cultural Comparison.

You will compare three magic squares: 1. **Lo Shu** (China, ~5th–1st c. BCE attested) 2. **Khajuraho** (India, ~10th–11th c. CE) 3. **Dürer** (Germany, 1514 CE)

Complete the table together. For each cell that asks about a sum, verify the sum yourself (don't take the question's word for it).

Comparison worksheet

Question	Lo Shu (3×3)	Khajuraho (4×4)	Dürer (4×4)
Order n	3	4	4
Magic constant	_____	_____	_____
Numbers used	1–_____	1–_____	1–_____
First row	, , _	, , ,	, , ,
Centre number(s)	_ (one cell)	, , , (four cells form the centre 2×2)	, , ,
Sum of corners	_ + _ + _ + _ = _	_ + _ + _ + _ = _	_ + _ + _ + _ = _
Every 2×2 panel sums to constant? (panel-magic?)	(irrelevant — no 2×2 panels in 3×3)	Yes / No	Yes / No
Approximate date	~5th–1st c. BCE	~10th–11th c. CE	1514 CE

Discussion (last 3 min)

- Which square is **most constrained** (i.e. satisfies the most rules)? Why?
- Which is **simplest** to construct? Which would you struggle with most?
- Do you see any reason the same idea appeared in three different places?

Reference data

Lo Shu (China):

```
4 | 9 | 2
3 | 5 | 7
8 | 1 | 6
```

Magic constant: 15

Khajuraho (India):

```
7 | 12 | 1 | 14
2 | 13 | 8 | 11
```

16		3		10		5
9		6		15		4

Magic constant: 34

Dürer (Germany):

16		3		2		13
5		10		11		8
9		6		7		12
4		15		14		1

Magic constant: 34

Answer key (teacher)

- Lo Shu magic constant: 15. Numbers 1–9.
- Khajuraho: 34. Numbers 1–16. **Panel-magic: Yes** (every 2×2 sums to 34).
- Dürer: 34. Numbers 1–16. **Panel-magic: Yes** (every 2×2 sums to 34).
- Both Khajuraho and Dürer are panel-magic; this is unusual.

Looks-like / sounds-like

	Looks like	Sounds like
Going well	Pair is summing cells in pencil, comparing answers	“Wait — let me check the top-left 2×2: 7+12+2+13 = 34 ✓”
Needs intervention	One partner fills in the worksheet; the other watches	Silence; “you do it”

Differentiation

- **Tier up:** Compare the four corners of each square. Do they sum to the magic constant? (Not in general, but yes for Dürer and Khajuraho — both are “associative” in this sense.)
- **Tier down:** Just verify the basic magic constant on one square. Skip the panel-magic question for now.

Project Brief

Project Brief — Module 14 Magic Squares (Middle)

Work sessions: Days 8 + 9 · **Presentation:** Day 10 · **Total marks:** 25 (project 20 + presentation 5)

The project

Each pair (or individual, with teacher approval) designs and presents an **original magic square** of order 5×5 or larger. Choose ONE of three formats:

Format 1: Labelled poster

Produce an A3-sized (or larger) poster that shows: - The constructed magic square (at least 5×5) - Step-by-step construction labels showing the *turagagati* placement order (number each cell 1, 2, 3, ... in the order the algorithm placed it) - A verification panel showing row, column, and diagonal sums - A 50-word caption stating the magic constant and citing Nārāyaṇa Paṇḍita's *Gaṇita-kaumudī*

Format 2: 60-second demo video

Make a short video that walks a Grade 5 viewer through constructing a 5×5 magic square. Constraints: - Total length: 60 seconds (± 10 sec) - Must show: the placement of at least cells 1, 2, 3, and one “blocked $\rightarrow$ directly below” step - Must include verbal verification of the magic constant - Cite the source verbally in the closing line (e.g. “Method from Nārāyaṇa Paṇḍita's *Gaṇita-kaumudī*, 1356 CE”)

Format 3: Written exposition

Produce a 2-page written piece that: - Constructs an original magic square (5×5 or larger) - Explains the construction in clear prose, step-by-step - Shows the verification arithmetic - Includes a paragraph on Nārāyaṇa Paṇḍita and the *Gaṇita-kaumudī* - Cites the source in a closing line

Design choices

You are not expected to invent a new algorithm. Your design choices are: - **Order (size)**. Pick 5×5 , 7×7 , or 9×9 . - **Starting cell**. The classical method starts in the top-middle. What happens if you start elsewhere? (You may use any valid starting cell — bottom-middle, top-right, etc. — and your magic square will be a rotation/reflection of the classical one.) - **Number range**. Default is 1 to n^2 . For tier-up, try a square with consecutive numbers from a different starting number (e.g. 10 to 34 for $n = 5$). Compute the new magic constant. - **Presentation format and style**. This is where most of your originality goes.

Why this matters

You're moving from *using* an algorithm to *teaching* an algorithm. Teaching is the proof of understanding. The verification step is what makes your work mathematics rather than craft.

What success looks like

A strong project will: - Demonstrate the *turagagati* method accurately (no arithmetic errors). - Be pitched for an audience (Grade 5 for the video, peer reader for poster/exposition). - Show your own design choices. - Include the verification panel — without exception.

Working in pairs

- Pair work is the default.
- Individual work allowed for any format with teacher approval.

- Groups of 3 acceptable — each member must have a distinct role (e.g. designer, constructor, verifier).

Citation rules

- At minimum **one** citation must appear visibly in the project.
- Format: “Nārāyaṇa Paṇḍita, *Gaṇita-kaumudī*, Book 14 (*Bhadragaṇita*), 1356 CE.”
- The source must be cited even if the artefact is a video — say it aloud at the end.

Self-reflection (submitted with project on Day 10)

In 4–6 sentences: 1. What did you set out to make, and what did you end up with? 2. What’s the most surprising thing you learned while making this? 3. If you had three more days, what would you do differently?